Robotica (2026), **xx**:xx 1–21
doi:10.1017/xxxxx

CAMBRIDGE
UNIVERSITY PRESS

RESEARCH ARTICLE

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Quadrotor Trajectory Tracking Under Unknown Wind Disturbances via Differential Flatness and Smooth Bounded Robust Control

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Received: xx xxx xxxx; **Revised:** xx xxx xxxx; **Accepted:** xx xxx xxxx

Abstract

Low-altitude observation over hilly terrain requires quadrotors to track manoeuvring trajectories while rejecting rapidly varying wind-force disturbances without saturated or oscillatory actuator commands. This paper proposes a disturbance-rejection framework that combines differential-flatness feedforward, smooth bounded feedback and disturbance-observer compensation. The feedforward channel generates predictable acceleration, thrust-direction and attitude commands from position and yaw derivatives, so that feedback mainly corrects residual errors. Wind-force disturbances, parametric uncertainty, unmodelled dynamics and virtual-control derivative terms are represented as equivalent disturbances. Both position and attitude loops use continuous $\tanh(\cdot)$-based bounded feedback, which gives gain-dependent bounds on correction and compensation terms. Regional uniform ultimate boundedness is established for an abstract second-order error system and its $\mathrm{SO}(3)$ attitude realization under smooth references, bounded disturbances and an attitude-error attraction-domain condition. MuJoCo simulations and real-flight tests evaluate manoeuvring tracking, wind rejection and input smoothness. In high-dynamic figure-eight tracking, the full controller reduces RMSE from $2.045\ \mathrm{m}$ without flatness feedforward to $0.021\ \mathrm{m}$. Under a synthetic spatiotemporal wind-force field, RMSE is reduced by 46.19% relative to DOB-PD and by 51.10% relative to PD. In wind-free hovering, integrated motor-thrust PSD over 0–$120\ \mathrm{Hz}$ is 49.09% lower than with SMC. A $30\ \mathrm{s}$ real-flight figure-eight window gives a position RMSE of $0.057\ \mathrm{m}$. These results indicate that separating predictable reference dynamics from unknown disturbances improves tracking accuracy and reduces broadband motor-input energy in the tested settings.

1. Introduction

Quadrotor unmanned aerial vehicles (UAVs) are widely used for terrain mapping, agricultural inspection, disaster monitoring and low-altitude environmental sensing because of their compact structure, vertical take-off and landing capability and high manoeuvrability[1, 2]. In near-ground flight over hilly terrain, wind rejection is only part of the challenge. The vehicle must also track manoeuvring trajectories, recover from disturbances and maintain smooth inputs under limited thrust and torque margins. Reference trajectories are often shaped jointly by terrain undulation, observation paths and obstacle-avoidance constraints, which introduce marked curvature variations and manoeuvring demands. Valley-channel acceleration, differential slope heating, terrain-induced flow and local vortex structures can also cause the wind field to vary rapidly with position, altitude and time[3, 4]. These disturbances are not simply constant biases in the translational channel. They affect position, attitude and equivalent aerodynamic parameters through underactuated coupling. Trajectory tracking under near-ground wind disturbances must therefore handle nonlinear coupling, model uncertainty, external disturbances and actuator constraints on both magnitude and smoothness.

Several modelling, estimation and control frameworks have been developed for quadrotor trajectory tracking[5, 6]. Differential flatness can algebraically generate nominal thrust, desired attitude and angular-velocity feedforward from position, yaw and their higher-order derivatives. It has therefore become an important tool for aggressive trajectory generation and tracking[7, 8, 9, 10]. High-speed flight studies have further incorporated aerodynamic effects, such as rotor drag, into flatness models to improve feedforward accuracy[11]. Thrust mixing, input-saturation handling and body-rate control have also improved actuator feasibility in agile flight[12, 13]. Time-optimal waypoint flight, online replanning and multi-fidelity optimization have extended manoeuvring capability on complex trajectories[14, 15, 16, 17, 18]. These studies provide a foundation for reference generation, feedforward compensation and actuator allocation. However, the coordination of flatness feedforward, disturbance compensation and continuous bounded feedback under persistent wind disturbances and model perturbations remains insufficiently resolved.

Robust control and disturbance compensation improve quadrotor disturbance rejection from the perspective of closed-loop correction. Backstepping control, sliding-mode control, adaptive control and barrier Lyapunov functions have been used to address underactuated coupling, external disturbances and state constraints[19, 20, 21, 22]. Geometric control on $\mathrm{SE}(3)$ or $\mathrm{SO}(3)$ avoids Euler-angle singularities and provides coordinate-free error descriptions for large-attitude manoeuvres[23, 24]. Disturbance observers, extended-state observers, uncertainty and disturbance estimators and incremental nonlinear dynamic inversion can compensate online for model mismatch, wind disturbances and unmodelled dynamics[25, 26, 27, 28, 29, 30, 31]. Work on wind sensing and aerodynamic disturbance estimation also indicates that practical wind-rejection performance depends on matching disturbance frequency characteristics, estimator structure, actuator bandwidth and thrust-generation mechanisms[32, 33, 34, 35].

The above studies have advanced trajectory feedforward, geometric feedback, disturbance estimation and actuator-constraint handling. Yet near-ground wind-disturbance missions still raise a practical question: which dynamics should be handled proactively by feedforward, which uncertain terms should be corrected online by feedback and observers, and how can these corrections remain continuous and bounded? Strong feedback correction and observer compensation help disturbance recovery, but thrust, torque and command rates are all limited by the actuation system. If robust or compensation terms are not smoothly bounded, control commands may contain spikes, chatter, actuator saturation or high-frequency components amplified by measurement noise. Conversely, relying only on saturation or excessive smoothing may reduce recovery capability under gusts and local strong disturbances. In meteorological observation over hilly terrain, vehicles often track spatially varying routes with limited thrust margins. The trade-off among accuracy, disturbance rejection and actuator feasibility is therefore central.

This paper proposes a quadrotor trajectory-tracking framework that combines differential-flatness feedforward, smooth bounded feedback and disturbance-observer compensation. The central idea is to separate predictable reference dynamics from bounded equivalent disturbances. Unlike studies focused mainly on flatness-based trajectory generation, flatness transformations and high-speed feedforward compensation[7, 10, 11], this work embeds flatness feedforward in a cascaded disturbance-rejection structure. The feedforward channel accounts for predictable high-order reference dynamics, so that feedback mainly corrects modelling errors, wind-force disturbances and discrete-implementation residuals. Wind-force disturbances, parametric perturbations, unmodelled dynamics, virtual-control derivative terms and feedforward implementation residuals are grouped into an equivalent total disturbance and compensated online by smooth disturbance observers. This treatment avoids the explicit expansion of high-order analytical derivatives in the implemented control law and keeps observer compensation

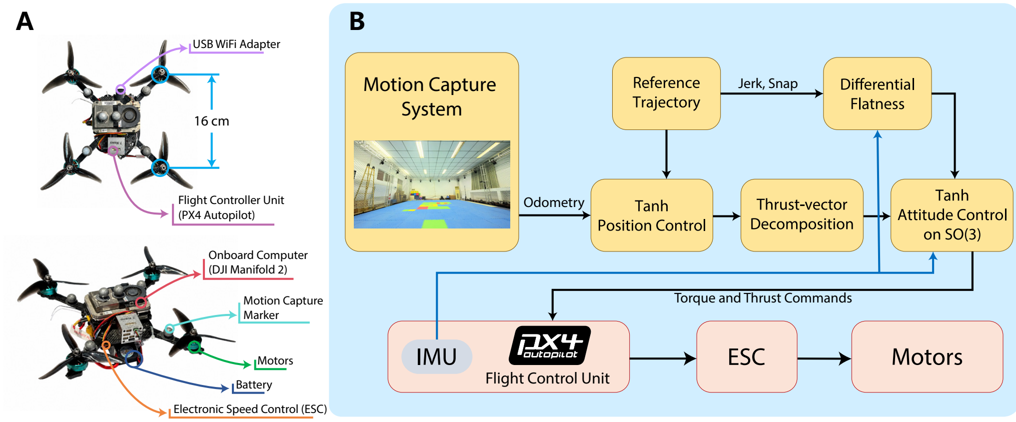

Figure 1. *Quadrotor platform and control framework: A, main hardware components of the quadrotor platform; B, overview of the framework implemented on the real UAV.*

consistent with the closed-loop boundedness analysis. Compared with studies emphasizing thrust mixing, actuator saturation or wind-disturbance estimation[12, 34], this paper uses $\tanh(\cdot)$-based smooth bounded feedback in both the position and attitude loops. The magnitudes of the error-correction and observer-compensation terms are bounded by design gain matrices, whereas the total thrust and torque still depend on reference derivatives, state rates, filtered estimates and actuator saturation. The attitude error is constructed directly on $\mathrm{SO}(3)$ to avoid singularities and representation ambiguities caused by local parametrizations. The overall control framework is shown in Fig. 1.

The main contributions of this paper are as follows:

1. A framework combining differential-flatness feedforward and smooth bounded disturbance-rejection feedback is proposed for near-ground trajectory tracking under wind-like force disturbances. It preserves the analytical generation capability of flatness feedforward while using bounded feedback and observer compensation to handle bounded force disturbances and modelling residuals.
2. A unified $\tanh(\cdot)$-type smooth feedback structure is constructed for the position and attitude loops. Virtual-control derivative terms are included in the equivalent total disturbance, avoiding explicit high-order analytical derivatives in the implemented controller. The magnitudes of the error-correction and disturbance-compensation terms are bounded by design gains and can therefore be coordinated with rotor thrust limits.
3. Regional uniform ultimate boundedness is established for an abstract second-order error system and its diagonal channel-wise application to the position and $\mathrm{SO}(3)$ attitude-error loops. The analysis clarifies how reference-trajectory smoothness, disturbance envelopes, observer-compensation magnitudes, feedforward residuals and attitude attraction domains affect the closed-loop guarantees.

The remainder of the paper is organized as follows. Section 2 presents the quadrotor dynamics, notation and control objective. Section 3 introduces flatness-based feedforward and desired-attitude generation. Section 4 constructs smooth bounded feedback and disturbance-observer compensation and provides the stability analysis. Section 5 maps the design to the position and attitude control laws. Sections 6 and 7 present simulation and real-flight validation, respectively. Section 8 concludes the paper.

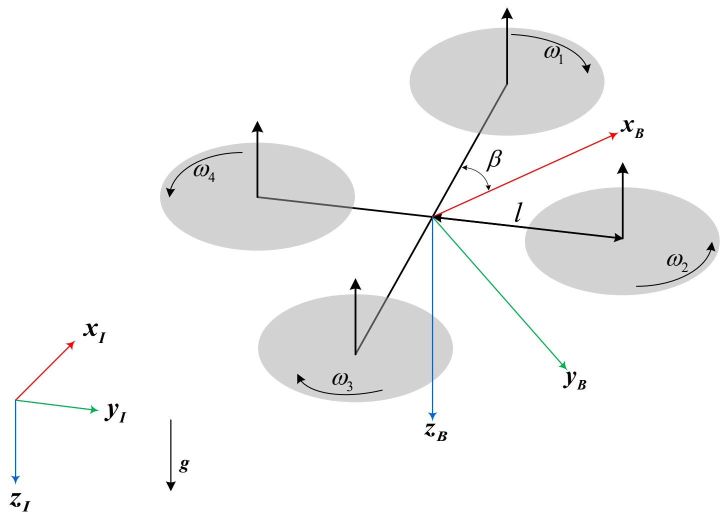

Figure 2. *Coordinate-frame definitions and rotor-numbering convention.*

2. Modelling and control objective

The following notation is used throughout the paper. Bold lowercase letters denote vectors, and bold uppercase letters denote matrices. The subscript r denotes the reference trajectory or a feedforward quantity generated from it, whereas the subscript d denotes a desired quantity sent to the attitude loop or actuator-allocation module. The function $\tanh(\cdot)$ acts elementwise on vectors. If $\boldsymbol{A}$ is a positive definite diagonal matrix, each component of $\boldsymbol{A}\tanh(\boldsymbol{Bx})$ is bounded by the corresponding diagonal element of $\boldsymbol{A}$.

Two right-handed coordinate frames are used. The inertial frame is $\mathcal{F}_I : \{\boldsymbol{x}_I, \boldsymbol{y}_I, \boldsymbol{z}_I\}$, where $\boldsymbol{z}_I$ points downward and is aligned with gravity. The body frame is $\mathcal{F}_B : \{\boldsymbol{x}_B, \boldsymbol{y}_B, \boldsymbol{z}_B\}$, where $\boldsymbol{x}_B$ points forward and $\boldsymbol{z}_B$ is opposite to the total-thrust direction, as shown in Fig. 2. Thus, in this NED convention, $\boldsymbol{g} = [0, 0, g]^T$ with $g > 0$. A vector with superscript B is expressed in the body frame; otherwise, it is expressed in the inertial frame. The rotation from $\mathcal{F}_B$ to $\mathcal{F}_I$ is denoted by $\boldsymbol{R} = [\boldsymbol{x}_B, \boldsymbol{y}_B, \boldsymbol{z}_B] \in \mathrm{SO}(3)$. For any $\boldsymbol{x} \in \mathbb{R}^3$, $\boldsymbol{x}^\wedge \in \mathfrak{so}(3)$ is the skew-symmetric matrix satisfying $\boldsymbol{x}^\wedge \boldsymbol{y} = \boldsymbol{x} \times \boldsymbol{y}$, and $(\cdot)^\vee$ is its inverse map.

The translational dynamics of the quadrotor are written as

$$\ddot{\boldsymbol{\xi}} = \frac{-T\boldsymbol{z}_B + \boldsymbol{f}_a}{m} + \boldsymbol{g}, \tag{1}$$

where $\boldsymbol{\xi}$ is the centre-of-mass position, m is the total mass, T is the scalar total thrust, $\boldsymbol{g}$ is the gravity vector and $\boldsymbol{f}_a$ denotes the equivalent external force caused by wind disturbances, parametric perturbations and unmodelled aerodynamic effects. The rotational kinematics and dynamics are

$$\dot{\boldsymbol{R}} = \boldsymbol{R}(\boldsymbol{\Omega}^B)^\wedge, \tag{2}$$

$$\boldsymbol{I}_v\dot{\boldsymbol{\Omega}}^B = -\boldsymbol{\Omega}^B \times (\boldsymbol{I}_v\boldsymbol{\Omega}^B) + \boldsymbol{\tau} + \boldsymbol{\tau}_{\mathrm{dis}}, \tag{3}$$

where $\boldsymbol{\Omega}^B$ is the body-frame angular velocity, $\boldsymbol{I}_v$ is the inertia matrix, $\boldsymbol{\tau}$ is the control torque generated by the rotors and $\boldsymbol{\tau}_{\mathrm{dis}}$ is the external disturbance torque.

To connect the rigid-body generalized inputs with the rotor actuators, define the rotor-speed vector $\boldsymbol{\omega} = [\omega_1, \omega_2, \omega_3, \omega_4]^T$ and the rotor-thrust vector $\boldsymbol{u} = [u_1, u_2, u_3, u_4]^T$. The thrust u_i and reaction-torque magnitude M_i of the ith rotor are given by the squared-speed model

$$u_i = k_F \omega_i^2, \qquad M_i = k_M \omega_i^2, \quad i = 1, \ldots, 4, \tag{4}$$

where k_F and k_M are the thrust and torque coefficients, respectively. Following the rotor numbering, arm length l and angle β in Fig. 2, the nominal control allocation is

$$\begin{bmatrix} T \\ \boldsymbol{\tau} \end{bmatrix} = \boldsymbol{B}_u \boldsymbol{u}, \qquad \boldsymbol{B}_u = \begin{bmatrix} 1 & 1 & 1 & 1 \\ l\sin\beta & -l\sin\beta & -l\sin\beta & l\sin\beta \\ l\cos\beta & l\cos\beta & -l\cos\beta & -l\cos\beta \\ -k_M/k_F & k_M/k_F & -k_M/k_F & k_M/k_F \end{bmatrix}. \tag{5}$$

Given the desired generalized control input $(T_d, \boldsymbol{\tau}_d)$ from the attitude loop, the unconstrained nominal inverse allocation is

$$\boldsymbol{u}_{\text{alloc}} = \boldsymbol{B}_u^{-1} \begin{bmatrix} T_d \\ \boldsymbol{\tau}_d \end{bmatrix}. \tag{6}$$

This allocation relation is used only for nominal input mapping. Actuator constraints are not embedded in the rigid-body dynamic model, and the actual rotor commands are generated in Section 5. Additional torques caused by rotor angular acceleration and gyroscopic effects are not included in the nominal allocation matrix. They are treated as bounded unmodelled terms and included in $\boldsymbol{\tau}_{\text{dis}}$.

Given a reference trajectory $\boldsymbol{\xi}_r(t)$ and reference yaw $\psi_r(t)$, the control objective is to keep the position error $\boldsymbol{\xi} - \boldsymbol{\xi}_r$, the velocity error $\dot{\boldsymbol{\xi}} - \dot{\boldsymbol{\xi}}_r$ and the attitude error on $\mathrm{SO}(3)$ bounded under bounded force disturbances and model uncertainty. These errors should enter tunable small neighbourhoods. At the same time, the feedback and observer-compensation signals should remain continuous and bounded by design gains, while the total thrust and torque are further constrained by reference derivatives, state rates, filtered estimates and actuator saturation.

3. Flatness-based feedforward

The proposed controller has feedforward and feedback channels. The feedforward channel uses the differential-flatness structure of the reference trajectory to generate nominal thrust, desired attitude, reference angular velocity and reference angular acceleration from $\boldsymbol{\xi}_r$, $\dot{\boldsymbol{\xi}}_r$, $\ddot{\boldsymbol{\xi}}_r$, $\boldsymbol{\xi}_r^{(3)}$, $\boldsymbol{\xi}_r^{(4)}$, ψ_r, $\dot{\psi}_r$ and $\ddot{\psi}_r$. The feedback channel uses a cascaded position and attitude structure to generate the desired thrust vector and desired torque. Predictable high-order reference dynamics are handled by the feedforward channel, whereas wind disturbances, parametric perturbations and unmodelled terms enter the smooth bounded feedback and observer-compensation design as equivalent disturbances.

The quadrotor can use the three-dimensional position $\boldsymbol{\xi}_r = [x_r, y_r, z_r]^T$ and yaw angle ψ_r as flat outputs. Given the reference acceleration $\ddot{\boldsymbol{\xi}}_r$, the nominal thrust vector without equivalent force disturbance is

$$\boldsymbol{F}_{\text{ff}} = m(\boldsymbol{g} - \ddot{\boldsymbol{\xi}}_r). \tag{7}$$

Let $\boldsymbol{F}_{\text{ff}}$ denote the nominal thrust vector along the body $\boldsymbol{z}_B$ axis. In the disturbance-free case with exact attitude tracking, it satisfies $T_r \boldsymbol{z}_{B,r} = \boldsymbol{F}_{\text{ff}}$. Thus,

$$T_r = \|\boldsymbol{F}_{\text{ff}}\|, \qquad \boldsymbol{z}_{B,r} = \frac{\boldsymbol{F}_{\text{ff}}}{\|\boldsymbol{F}_{\text{ff}}\|}, \qquad T_r > 0. \tag{8}$$

An intermediate axis is then constructed from the reference yaw angle as

$$\boldsymbol{x}_{C,r} = [\cos\psi_r,\ \sin\psi_r,\ 0]^T, \tag{9}$$

which gives

$$\boldsymbol{y}_{B,r} = \frac{\boldsymbol{z}_{B,r} \times \boldsymbol{x}_{C,r}}{\|\boldsymbol{z}_{B,r} \times \boldsymbol{x}_{C,r}\|}, \tag{10}$$
$$\boldsymbol{x}_{B,r} = \frac{\boldsymbol{y}_{B,r} \times \boldsymbol{z}_{B,r}}{\|\boldsymbol{y}_{B,r} \times \boldsymbol{z}_{B,r}\|}, \tag{11}$$
$$\boldsymbol{R}_r = [\boldsymbol{x}_{B,r}, \boldsymbol{y}_{B,r}, \boldsymbol{z}_{B,r}]. \tag{12}$$

Equation (10) requires $\boldsymbol{z}_{B,r}$ not to be parallel to $\boldsymbol{x}_{C,r}$. This condition means that the desired thrust direction must not be degenerate or collinear with the prescribed yaw axis. In implementation, if this degeneracy is approached, the previous feasible yaw axis or another orthogonal fallback axis is used.

If the reference trajectory also provides jerk $\boldsymbol{\xi}_r^{(3)}$ and snap $\boldsymbol{\xi}_r^{(4)}$, the reference angular velocity and angular acceleration required by the attitude loop can be generated. Define the computational attitude quantities

$$\boldsymbol{R}_c = [\boldsymbol{x}_{B,c}, \boldsymbol{y}_{B,c}, \boldsymbol{z}_{B,c}], \qquad T_c > 0, \qquad \boldsymbol{\Omega}_c = \boldsymbol{R}_c \boldsymbol{\Omega}_c^B. \tag{13}$$

In implementation, one may set $\boldsymbol{R}_c = \boldsymbol{R}$ and $\boldsymbol{\Omega}_c^B = \boldsymbol{\Omega}^B$, and obtain T_c from rotor-speed or thrust estimates. Alternatively, $\boldsymbol{R}_c = \boldsymbol{R}_r$, $\boldsymbol{\Omega}_c^B = \boldsymbol{\Omega}_r^B$ and $T_c = T_r$ can be used for strict reference feedforward. The former choice updates the flatness feedforward quantities with the current flight state. In this case, $\boldsymbol{\Omega}_d^B$ and $\boldsymbol{\alpha}_d^B$ are interpreted as current-state-corrected feedforward quantities rather than exact derivatives of the desired attitude trajectory $\boldsymbol{R}_d(t)$. The discrepancy between these implemented feedforward quantities and the strict $\boldsymbol{R}_d(t)$ derivatives is treated as a bounded feedforward residual in the attitude equivalent disturbance.

After replacing the thrust axis, angular velocity and thrust magnitude in the flatness relation $T\boldsymbol{z}_B = m(\boldsymbol{g} - \ddot{\boldsymbol{\xi}}_r)$ with the computational quantities $(\boldsymbol{z}_{B,c}, \boldsymbol{\Omega}_c, T_c)$, the reference jerk gives

$$\dot{T}_c = -m(\boldsymbol{\xi}_r^{(3)})^T \boldsymbol{z}_{B,c}, \tag{14}$$

$$\boldsymbol{h}_\Omega = \boldsymbol{\Omega}_d \times \boldsymbol{z}_{B,c} = -\frac{m}{T_c}\boldsymbol{\xi}_r^{(3)} - \frac{\dot{T}_c}{T_c}\boldsymbol{z}_{B,c}. \tag{15}$$

The feedforward component of the desired angular velocity in the body frame is therefore

$$\boldsymbol{\Omega}_d^B = \begin{bmatrix} -\boldsymbol{h}_\Omega^T \boldsymbol{y}_{B,c} \\ \boldsymbol{h}_\Omega^T \boldsymbol{x}_{B,c} \\ \dot{\psi}_r \boldsymbol{z}_I^T \boldsymbol{z}_{B,c} \end{bmatrix}. \tag{16}$$

Using the reference snap gives

$$\ddot{T}_c = -m(\boldsymbol{\xi}_r^{(4)})^T \boldsymbol{z}_{B,c} - m(\boldsymbol{\xi}_r^{(3)})^T(\boldsymbol{\Omega}_c \times \boldsymbol{z}_{B,c}), \tag{17}$$

$$\begin{aligned} \boldsymbol{h}_\alpha &= \boldsymbol{\alpha}_d \times \boldsymbol{z}_{B,c} \\ &= -\frac{m}{T_c}\boldsymbol{\xi}_r^{(4)} - \frac{2\dot{T}_c}{T_c}(\boldsymbol{\Omega}_c \times \boldsymbol{z}_{B,c}) - \frac{\ddot{T}_c}{T_c}\boldsymbol{z}_{B,c} - \boldsymbol{\Omega}_c \times (\boldsymbol{\Omega}_c \times \boldsymbol{z}_{B,c}). \end{aligned} \tag{18}$$

The feedforward component of the desired angular acceleration in the body frame is then

$$\boldsymbol{\alpha}_d^B = \begin{bmatrix} -\boldsymbol{h}_\alpha^T \boldsymbol{y}_{B,c} \\ \boldsymbol{h}_\alpha^T \boldsymbol{x}_{B,c} \\ \ddot{\psi}_r \boldsymbol{z}_I^T \boldsymbol{z}_{B,c} \end{bmatrix}. \tag{19}$$

The third component of the yaw angular acceleration in Eq. (19) uses a common approximation. The complete derivative also contains a coupling residual caused by variation in the thrust axis:

$$\delta_\psi = \dot{\psi}_r \boldsymbol{z}_I^T (\boldsymbol{\Omega}_c \times \boldsymbol{z}_{B,c}), \qquad |\delta_\psi| \le |\dot{\psi}_r| \, \|\boldsymbol{\Omega}_c\|. \tag{20}$$

When the reference yaw rate and the thrust-axis variation rate are bounded, this residual is included in the attitude equivalent-disturbance envelope.

More generally, let $\boldsymbol{r}_{\mathrm{ff},\Omega}$ denote the residual caused by using the current-state-corrected flatness feedforward instead of the exact time derivatives of $\boldsymbol{R}_d(t)$. The attitude-loop analysis assumes that this residual is bounded channel by channel in the considered operating region:

$$|r_{\mathrm{ff},\Omega,j}(t)| \le \bar{r}_{\mathrm{ff},\Omega,j}, \qquad j = 1, 2, 3. \tag{21}$$

4. Smooth bounded feedback and stability analysis

To provide a common theoretical basis for the position and attitude loops, this section first considers a second-order error system with unmatched and matched disturbances:

$$\begin{cases} \dot{x}_1 = x_2 + d_1(t), \\ \dot{x}_2 = f(\boldsymbol{x}) + d_2(t) + v, \end{cases} \tag{22}$$

where $\boldsymbol{x} = (x_1, x_2)^T$ is the measurable state, v is the control input, $d_1(t)$ is an unmatched disturbance and $d_2(t)$ is a matched disturbance satisfying

$$|d_i(t)| \le D_i, \qquad i = 1, 2. \tag{23}$$

The variable x_1 represents the output error, and x_2 represents its first-order error channel. Because $d_1(t)$ may persist, the control objective is practical tracking with a prescribed accuracy:

$$|x_1(t)| \le \delta_1, \qquad t \ge t_1. \tag{24}$$

The smooth bounded virtual control is defined as

$$e_1 = x_1, \qquad e_2 = x_2 + m_1 \tanh(k_1 e_1), \tag{25}$$

where $m_1 > 0$ and $k_1 > 0$. The error system is then

$$\begin{cases} \dot{e}_1 = e_2 + d_1(t) - m_1 \tanh(k_1 e_1), \\ \dot{e}_2 = f(\boldsymbol{x}) + v + \tilde{d}_2(t), \end{cases} \tag{26}$$

where

$$\tilde{d}_2(t) = d_2(t) + \Lambda(t), \qquad \Lambda(t) = m_1 k_1 (1 - \tanh^2(k_1 e_1)) \dot{e}_1. \tag{27}$$

Because $\Lambda(t)$ contains $\dot{e}_1$, its global boundedness cannot be assumed before boundedness of e_2 is established. From $|\tanh(\cdot)| \leq 1$, the following state-dependent estimate is obtained:

$$|\Lambda(t)| \leq m_1 k_1 (|e_2(t)| + D_1 + m_1). \tag{28}$$

The resulting control law is

$$v = -\left(f(\boldsymbol{x}) + m_2 \tanh(k_2 e_2) + k_3 \dot{x}_2\right), \tag{29}$$

where $m_2 > 0$, $k_2 > 0$ and $k_3 \geq 0$. Combining this expression with $\dot{x}_2 = \dot{e}_2 - \Lambda(t)$, define

$$\Delta_2(t) = d_2(t) + (1 + k_3)\Lambda(t), \tag{30}$$

and the closed-loop system becomes

$$\begin{cases} \dot{e}_1 = e_2 + d_1(t) - m_1 \tanh(k_1 e_1), \\ \dot{e}_2 = \dfrac{-m_2 \tanh(k_2 e_2) + \Delta_2(t)}{1 + k_3}. \end{cases} \tag{31}$$

Proposition 1. Given $\delta_1 > 0$, $\delta_2 > 0$ and $R_2 > \delta_2$, suppose that system (22) satisfies (23). Then, for any $k_3 \geq 0$, one can choose

$$k_1 \geq 1/\delta_1, \qquad k_2 \geq 1/\delta_2, \tag{32}$$

$$m_1 > \frac{R_2 + D_1}{0.76}, \tag{33}$$

$$m_2 > \frac{D_2 + (1 + k_3) m_1 k_1 (R_2 + D_1 + m_1)}{0.76}, \tag{34}$$

such that all closed-loop trajectories satisfying $|e_2(0)| \leq R_2$ remain in $|e_2(t)| \leq R_2$ and enter $|e_2(t)| \leq \delta_2$ after a finite time. The error e_1 also remains bounded and enters $|e_1(t)| \leq \delta_1$ after a finite time.

Proof. Let $V_2 = 0.5 e_2^2$. When $1/k_2 < |e_2| \leq R_2$, $|\tanh(k_2 e_2)| > 0.76$. From (31) and (28),

$$\dot{V}_2 \leq \frac{|e_2|}{1 + k_3} \left[-0.76 m_2 + D_2 + (1 + k_3) m_1 k_1 (R_2 + D_1 + m_1)\right]. \tag{35}$$

Condition (34) ensures $\dot{V}_2 < 0$ in the annular region $1/k_2 < |e_2| \leq R_2$. Hence, the set $|e_2| \leq R_2$ is positively invariant, and the trajectory enters $|e_2| \leq 1/k_2 \leq \delta_2$. Before e_2 enters the δ_2 neighbourhood, positive invariance still gives $|e_2(t)| \leq R_2$. In addition, $|\tanh(\cdot)| \leq 1$ gives

$$|\dot{e}_1| \leq R_2 + D_1 + m_1, \tag{36}$$

Therefore, e_1 is bounded on any finite transient interval. Let $V_1 = 0.5 e_1^2$. When $|e_1| > 1/k_1$, $|\tanh(k_1 e_1)| > 0.76$. Together with $|e_2(t)| \leq R_2$, this gives

$$\dot{V}_1 \leq |e_1| (R_2 + D_1 - 0.76 m_1). \tag{37}$$

By (33), $\dot{V}_1 < 0$. Hence, e_1 cannot leave the compact set determined by $e_1(0)$ and $1/k_1$, and it enters $|e_1| \leq 1/k_1 \leq \delta_1$ after a finite time. This completes the proof.

The disturbance observer uses the same smooth bounded correction idea. Consider the scalar observer error

$$\dot{\varepsilon} = d_{\mathrm{eq}}(t) - P \tanh(L\varepsilon), \tag{38}$$

where $L > 0$ and $|d_{\mathrm{eq}}(t)| \le \bar{d}$. If $P > \bar{d}$, then for any $\eta \in (0, P - \bar{d})$, the observer error enters and remains in the neighbourhood

$$r_\varepsilon = \frac{1}{L}\,\mathrm{artanh}\left(\frac{\bar{d}+\eta}{P}\right) \tag{39}$$

defined in (39). This result follows directly from $V_\varepsilon = 0.5\varepsilon^2$, because when $|\varepsilon| > r_\varepsilon$,

$$\dot{V}_\varepsilon \le |\varepsilon|\left(\bar{d} - P|\tanh(L\varepsilon)|\right) < -\eta|\varepsilon|. \tag{40}$$

In the vector case, if $\boldsymbol{P}$ and $\boldsymbol{L}$ are positive definite diagonal matrices and each channel satisfies $P_j > \bar{d}_j$, the above conclusion holds channel by channel. The observer guarantees convergence of the error to a computable neighbourhood rather than exact pointwise reconstruction of the true disturbance. Residual observer error, measurement noise and implementation error still enter the disturbance bound in Proposition 1 as closed-loop equivalent disturbances.

Corollary 1. Consider an n-channel error system with diagonal gain matrices. Suppose that, in a compact operating region $\mathcal{D}$, the jth channel can be written as

$$\dot{e}_{1j} = e_{2j} + d_{1j}(t) - m_{1j}\tanh(k_{1j}e_{1j}), \tag{41}$$

$$\dot{e}_{2j} = \frac{-m_{2j}\tanh(k_{2j}e_{2j}) + \Delta_{2j}(t)}{1 + k_{3j}}, \qquad j = 1, \ldots, n, \tag{42}$$

where $|d_{1j}(t)| \le D_{1j}$ and $|\Delta_{2j}(t)| \le \bar{D}_{2j}$. Proposition 1 then applies to each channel separately. Hence, for prescribed δ_{1j} and δ_{2j}, suitable diagonal gains ensure finite-time entry into $|e_{1j}| \le \delta_{1j}$ and $|e_{2j}| \le \delta_{2j}$, provided the trajectory remains in $\mathcal{D}$. Consequently,

$$\|\boldsymbol{e}_1(t)\| \le \left(\sum_{j=1}^{n}\delta_{1j}^2\right)^{1/2}, \qquad \|\boldsymbol{e}_2(t)\| \le \left(\sum_{j=1}^{n}\delta_{2j}^2\right)^{1/2} \tag{43}$$

after a finite time. Cross-channel coupling, geometric residuals, feedforward mismatch, observer residuals and actuator-allocation residuals are not assumed to vanish; they must be included in the channel disturbance envelopes D_{1j} and $\bar{D}_{2j}$.

5. Quadrotor control laws

5.1. Position loop

Let the desired position be $\boldsymbol{\xi}_r$. Define the position error and the velocity-type composite error as

$$\boldsymbol{e}_\xi = \boldsymbol{\xi} - \boldsymbol{\xi}_r, \tag{44}$$

$$\boldsymbol{e}_v = \dot{\boldsymbol{\xi}} - \dot{\boldsymbol{\xi}}_r + \boldsymbol{M}_\xi\tanh(\boldsymbol{K}_\xi\boldsymbol{e}_\xi). \tag{45}$$

The position control law combines flatness-based feedforward acceleration, smooth error feedback and disturbance compensation in the desired thrust vector. To avoid an algebraic loop between the actual acceleration and the control input, the control law uses the estimated linear acceleration $\hat{\boldsymbol{a}}$:

$$T_d z_{B,d} = m\left(\boldsymbol{g} - \ddot{\boldsymbol{\xi}}_r + \boldsymbol{M}_v\tanh(\boldsymbol{K}_v\boldsymbol{e}_v) + \boldsymbol{K}_a(\hat{\boldsymbol{a}} - \ddot{\boldsymbol{\xi}}_r) + \boldsymbol{\mu}_v\right). \tag{46}$$

where $\boldsymbol{\mu}_\nu$ is the output of the position disturbance observer. From (1) and (45), the total disturbance in the velocity-type composite-error dynamics can be written as

$$\tilde{\boldsymbol{f}} = \frac{\boldsymbol{f}_a}{m} + \frac{d}{dt}\left(\boldsymbol{M}_\xi \tanh(\boldsymbol{K}_\xi \boldsymbol{e}_\xi)\right), \tag{47}$$

where the first term corresponds to external wind disturbances and model uncertainty, and the second term corresponds to the derivative of the backstepping virtual control. This derivative term is not expanded in the implemented control law. Instead, it is included with the wind disturbance in the equivalent total disturbance.

The position disturbance observer is

$$\dot{\hat{\boldsymbol{e}}}_\nu = -\frac{T_d \boldsymbol{z}_{B,d}}{m} + \boldsymbol{g} - \ddot{\boldsymbol{\xi}}_r + \boldsymbol{\mu}_\nu, \tag{48}$$

with

$$\boldsymbol{\varepsilon}_\nu = \boldsymbol{e}_\nu - \hat{\boldsymbol{e}}_\nu, \qquad \boldsymbol{\mu}_\nu = \boldsymbol{P}_\nu \tanh(\boldsymbol{L}_\nu \boldsymbol{\varepsilon}_\nu). \tag{49}$$

If $|d_{\nu,\mathrm{eq},j}(t)| \le \bar{d}_{\nu,j}$ and $P_{\nu,j} > \bar{d}_{\nu,j}$ hold channel by channel in a given operating region, the observer error enters the neighbourhood given by (39), and the compensation magnitude is bounded by $\boldsymbol{P}_\nu$.

5.2. Thrust-vector decomposition

From (46), the desired total-thrust magnitude is

$$T_d = \left\| m\left(\boldsymbol{g} - \ddot{\boldsymbol{\xi}}_r + \boldsymbol{M}_\nu \tanh(\boldsymbol{K}_\nu \boldsymbol{e}_\nu) + \boldsymbol{K}_a(\hat{\boldsymbol{a}} - \ddot{\boldsymbol{\xi}}_r) + \boldsymbol{\mu}_\nu\right)\right\|. \tag{50}$$

The desired thrust-axis direction is

$$\boldsymbol{z}_{B,d} = \frac{\boldsymbol{g} - \ddot{\boldsymbol{\xi}}_r + \boldsymbol{M}_\nu \tanh(\boldsymbol{K}_\nu \boldsymbol{e}_\nu) + \boldsymbol{K}_a(\hat{\boldsymbol{a}} - \ddot{\boldsymbol{\xi}}_r) + \boldsymbol{\mu}_\nu}{\left\|\boldsymbol{g} - \ddot{\boldsymbol{\xi}}_r + \boldsymbol{M}_\nu \tanh(\boldsymbol{K}_\nu \boldsymbol{e}_\nu) + \boldsymbol{K}_a(\hat{\boldsymbol{a}} - \ddot{\boldsymbol{\xi}}_r) + \boldsymbol{\mu}_\nu\right\|}. \tag{51}$$

The reference yaw angle then gives

$$\boldsymbol{x}_{C,d} = [\cos\psi_r,\ \sin\psi_r,\ 0]^T, \tag{52}$$

and

$$\boldsymbol{y}_{B,d} = \frac{\boldsymbol{z}_{B,d} \times \boldsymbol{x}_{C,d}}{\|\boldsymbol{z}_{B,d} \times \boldsymbol{x}_{C,d}\|}, \tag{53}$$

$$\boldsymbol{x}_{B,d} = \frac{\boldsymbol{y}_{B,d} \times \boldsymbol{z}_{B,d}}{\|\boldsymbol{y}_{B,d} \times \boldsymbol{z}_{B,d}\|}, \tag{54}$$

$$\boldsymbol{R}_d = [\boldsymbol{x}_{B,d}, \boldsymbol{y}_{B,d}, \boldsymbol{z}_{B,d}]. \tag{55}$$

If reference jerk and snap are available, $\boldsymbol{\Omega}_d^B$ and $\boldsymbol{\alpha}_d^B$ are generated according to Section 3. If these high-order reference quantities are unavailable, one may set $\boldsymbol{\Omega}_d^B = \boldsymbol{0}$ and $\boldsymbol{\alpha}_d^B = \boldsymbol{0}$. The controller then reduces to an attitude-reference plus smooth-feedback structure.

5.3. Attitude loop on SO(3)

The attitude error is defined directly on $\mathrm{SO}(3)$. Following the classical construction in geometric control[23, 24], define the attitude-error function

$$\Psi(\boldsymbol{R}, \boldsymbol{R}_d) = \frac{1}{2}\mathrm{tr}\left(\boldsymbol{I} - \boldsymbol{R}_d^T \boldsymbol{R}\right), \tag{56}$$

and the geometric attitude-error vector

$$\boldsymbol{e}_R = \frac{1}{2}\left(\boldsymbol{R}_d^T \boldsymbol{R} - \boldsymbol{R}^T \boldsymbol{R}_d\right)^{\vee}. \tag{57}$$

The angular-velocity error is

$$\boldsymbol{e}_{\Omega,g} = \boldsymbol{\Omega}^B - \boldsymbol{R}^T \boldsymbol{R}_d \boldsymbol{\Omega}_d^B. \tag{58}$$

On this basis, construct the angular-velocity-type composite error

$$\boldsymbol{e}_\Omega = \boldsymbol{e}_{\Omega,g} + \boldsymbol{M}_R \tanh(\boldsymbol{K}_R \boldsymbol{e}_R). \tag{59}$$

When $\Psi < \psi_{\max} < 2$, $\boldsymbol{e}_R$ is equivalent to the local attitude error, and the error map is bounded. Within this attraction domain, the attitude loop can therefore be treated channel by channel as a second-order error system with bounded geometric coupling terms. These coupling terms, the yaw-feedforward residual in Eq. (20), the current-state feedforward residual in Eq. (21), observer residuals and actuator-allocation residuals are included in the attitude equivalent-disturbance envelope.

The desired torque control law is

$$\begin{aligned}\boldsymbol{\tau}_d = {} & \boldsymbol{\Omega}^B \times (\boldsymbol{I}_v \boldsymbol{\Omega}^B) - \boldsymbol{I}_v \boldsymbol{\mu}_\Omega \\ & - \boldsymbol{I}_v \left((\boldsymbol{\Omega}^B)^\wedge \boldsymbol{R}^T \boldsymbol{R}_d \boldsymbol{\Omega}_d^B - \boldsymbol{R}^T \boldsymbol{R}_d \boldsymbol{\alpha}_d^B\right) \\ & - \boldsymbol{I}_v \boldsymbol{M}_\Omega \tanh(\boldsymbol{K}_\Omega \boldsymbol{e}_\Omega) - \boldsymbol{I}_v \boldsymbol{K}_\alpha \widehat{\dot{\boldsymbol{e}}}_{\Omega,g}.\end{aligned} \tag{60}$$

where $\widehat{\dot{\boldsymbol{e}}}_{\Omega,g}$ can be obtained from angular-acceleration estimation, filtered differentiation or a state estimator, rather than by instantaneous inversion of the torque equation. The corresponding estimation and filtering residual is included in the attitude equivalent disturbance.

Given $(T_d, \boldsymbol{\tau}_d)$, the actuator layer computes the nominal rotor thrusts $\boldsymbol{u}_{\mathrm{alloc}}$ according to (6) and then generates commands satisfying the per-rotor thrust range:

$$\boldsymbol{u}_{\mathrm{cmd}} = \mathrm{sat}_{[0,u_{\max}]}(\boldsymbol{u}_{\mathrm{alloc}}), \qquad \omega_{i,d} = \sqrt{\frac{u_{\mathrm{cmd},i}}{k_F}}. \tag{61}$$

where $\mathrm{sat}_{[0,u_{\max}]}(\cdot)$ denotes the per-rotor thrust constraint in the implementation layer. The generalized input residual caused by saturation is

$$\boldsymbol{\delta}_{T\tau} = \boldsymbol{B}_u(\boldsymbol{u}_{\mathrm{cmd}} - \boldsymbol{u}_{\mathrm{alloc}}) = [\delta_T, \boldsymbol{\delta}_\tau^T]^T. \tag{62}$$

where δ_T and the attitude-tracking error together form the thrust-vector implementation error and are included in the equivalent disturbance of the position loop. The term $\boldsymbol{\delta}_\tau$ is included in the equivalent disturbance of the attitude loop. This residual does not change the nominal rigid-body model in Section 2, but it enters the closed-loop disturbance envelope as a bounded actuator residual.

Table 1. *Simulated quadrotor parameters.*

Parameter	Symbol	Value
Mass	m	0.81 kg
Body inertia	$\mathrm{diag}(\boldsymbol{I}_v)$	$(1.91, 2.37, 3.60) \times 10^{-3}\ \mathrm{kg\,m^2}$
Rotor position in the body frame	$\boldsymbol{r}_i$	$(\pm 0.09, \pm 0.09, 0.05)$ m
Single-rotor thrust command range	u_i	$[0, 14.80]$ N
Rotor torque-to-thrust coefficient ratio	k_M/k_F	0.05

The attitude disturbance observer is

$$\begin{aligned}\dot{\hat{\boldsymbol{e}}}_\Omega = &-\boldsymbol{I}_v^{-1}(\boldsymbol{\Omega}^B \times (\boldsymbol{I}_v\boldsymbol{\Omega}^B)) + \boldsymbol{I}_v^{-1}\boldsymbol{\tau}_d \\ &+ (\boldsymbol{\Omega}^B)^\wedge \boldsymbol{R}^T\boldsymbol{R}_d\boldsymbol{\Omega}_d^B - \boldsymbol{R}^T\boldsymbol{R}_d\boldsymbol{\alpha}_d^B + \boldsymbol{\mu}_\Omega.\end{aligned} \tag{63}$$

with

$$\boldsymbol{\varepsilon}_\Omega = \boldsymbol{e}_\Omega - \hat{\boldsymbol{e}}_\Omega, \qquad \boldsymbol{\mu}_\Omega = \boldsymbol{P}_\Omega \tanh(\boldsymbol{L}_\Omega\boldsymbol{\varepsilon}_\Omega). \tag{64}$$

If $|d_{\Omega,\mathrm{eq},j}(t)| \le \bar{d}_{\Omega,j}$ and $P_{\Omega,j} > \bar{d}_{\Omega,j}$ hold channel by channel in a given operating region, the attitude observer error enters a computable neighbourhood, and the magnitude of $\boldsymbol{\mu}_\Omega$ is bounded by $\boldsymbol{P}_\Omega$.

Remark 1. The above closed-loop conclusion for the quadrotor is regional and channel-wise. Together with Proposition 1 and Corollary 1, it indicates that the closed-loop position and attitude errors remain regionally uniformly ultimately bounded for a given compact set of initial errors if the following conditions hold: the reference trajectory is sufficiently smooth, the desired thrust satisfies $T_d > 0$, the initial attitude error remains in the attraction domain $\Psi < \psi_{\max} < 2$, the position and attitude equivalent disturbances are bounded channel by channel, and the observer-compensation magnitudes and feedback gains cover the residual disturbance envelopes. The equivalent disturbance envelopes include cross-channel coupling, geometric residuals, the feedforward mismatch in (21), estimation and filtering residuals, and the thrust and torque residuals generated by per-rotor saturation through (62). Thus, the analysis neither requires perfect attitude-loop tracking nor equates actuator feasibility with boundedness of the $\tanh(\cdot)$ feedback terms alone.

6. Simulations

The simulations address three linked questions: whether flatness feedforward captures the manoeuvring reference dynamics, whether disturbance observation and smooth bounded feedback improve tracking under a synthetic spatiotemporal wind disturbance, and whether the continuous bounded structure reduces broadband oscillations in motor inputs. The three experiments use the same rigid-body model, per-rotor thrust limits and sampling settings. Within each experiment, all compared controllers use the same initial state, target trajectory and disturbance input.

6.1. Simulation setup

The simulation platform uses MuJoCo [36] to construct the quadrotor rigid-body model. The gravity direction, thrust-axis definition and control-input interface are consistent with the model in Section 2. The simulation step is 0.001 s. The position and attitude loops run at 100 Hz and 800 Hz, respectively. The quadrotor parameters are listed in Table 1.

Table 2 gives the controller parameters used in the simulations. Position vectors are ordered as (x, y, z), and attitude vectors are ordered as (ϕ, θ, ψ). In the proposed controller, M and K denote the magnitude and slope of the $\tanh(\cdot)$ feedback terms, respectively. In PD and DOB-PD, λ denotes the linear feedback slope. In SMC, η and φ denote the saturation magnitude and boundary-layer width, respectively. Observer outputs are filtered by a first-order low-pass filter. The cutoff frequencies of the position and attitude loops are 10 Hz and 25 Hz for both the proposed controller and DOB-PD. The comparison-controller gains were tuned manually under the same sampling rates, sensor noise levels and per-rotor thrust limits, with the objective of reducing tracking RMSE while avoiding sustained actuator saturation or visually evident divergence. The reported comparisons should therefore be interpreted as controlled implementation comparisons rather than exhaustive optimal tuning of each baseline.

The initial position in all experiments is $(0, 0, -0.3)$ m in the NED coordinate system. The trajectory reference altitude and hovering target altitude are both $z = -1.0$ m. All controllers share the per-rotor thrust range in Table 1. The measurement update frequencies of position, velocity, attitude and angular velocity are 30, 100, 400 and 800 Hz, respectively. The position-error feedback term is updated at the position-measurement frequency. Measurement noise is zero-mean additive noise, and the same noise sequence is used within each comparison. The standard deviations of position, velocity, linear acceleration, attitude, angular velocity and angular acceleration are 0.002 m, 0.01 m/s, $0.05\ \mathrm{m\,s^{-2}}$, 0.0015 rad, $0.005\ \mathrm{rad\,s^{-1}}$ and $0.05\ \mathrm{rad\,s^{-2}}$, respectively.

The hovering input-smoothness analysis uses equivalent motor thrusts recorded at approximately 800 Hz, with an analysis-window length of 105 s. The PSD is computed using a 1024-point Hann window with 896-point overlap between adjacent windows. The spectra are evaluated in linear units before integration, so the plotted PSD has units proportional to $\mathrm{N^2/Hz}$ and the integrated band energy has units of $\mathrm{N^2}$. Frequency-band energy is obtained by integrating the linear PSD over 0–120, 20–120, 40–120 and 80–120 Hz.

Table 2. *Controller parameters used in the simulations.*

Controller	Position-loop parameters	Attitude-loop parameters	Observer gains
Proposed	$M_\xi = (1.65, 1.65, 0.80)$	$M_R = (0.50, 0.50, 1.00)$	$P_v = (1.50, 1.50, 4.20)$
	$K_\xi = (1.35, 1.35, 7.00)$	$K_R = (30.0, 30.0, 10.0)$	$L_v = (1.00, 1.00, 2.50)$
	$M_v = (4.00, 4.00, 6.00)$	$M_\Omega = (20.0, 20.0, 15.0)$	$P_\Omega = (10.0, 10.0, 10.0)$
	$K_v = (4.00, 4.00, 1.00)$	$K_\Omega = (10.0, 10.0, 3.0)$	$L_\Omega = (3.50, 3.50, 2.00)$
	$K_a = (1.00, 1.00, 1.00)$	$K_\alpha = (1.00, 1.00, 0.20)$	
PD	linear slope	linear slope	none
	$\lambda_\xi = (0.960, 0.960, 3.850)$	$\lambda_R = (12.0, 12.0, 8.0)$	
	$\lambda_v = (3.000, 3.000, 3.600)$	$\lambda_\Omega = (126.0, 126.0, 30.0)$	
	$K_a = (0.40, 0.40, 0.80)$	$K_\alpha = (0.70, 0.70, 0.20)$	
DOB-PD	linear slope	linear slope	$P_v = (1.20, 1.20, 3.00)$
	$\lambda_\xi = (1.013, 1.013, 3.850)$	$\lambda_R = (12.0, 12.0, 8.0)$	$L_v = (1.00, 1.00, 2.00)$
	$\lambda_v = (2.160, 2.160, 3.600)$	$\lambda_\Omega = (126.0, 126.0, 30.0)$	$P_\Omega = (5.00, 5.00, 8.00)$
	$K_a = (0.40, 0.40, 0.80)$	$K_\alpha = (0.70, 0.70, 0.20)$	$L_\Omega = (2.00, 2.00, 2.00)$
SMC	$\eta_\xi = (1.40, 1.40, 0.80)$	$\eta_R = (0.80, 0.80, 1.00)$	none
	$\varphi_\xi = (0.500, 0.500, 0.125)$	$\varphi_R = (0.025, 0.025, 0.083)$	
	$\eta_v = (3.50, 3.50, 5.50)$	$\eta_\Omega = (26.0, 26.0, 15.0)$	
	$\varphi_v = (0.200, 0.200, 0.500)$	$\varphi_\Omega = (0.045, 0.045, 0.200)$	
	$K_a = (0.90, 0.90, 1.00)$	$K_\alpha = (0.90, 0.90, 0.30)$	

6.2. *High-maneuver tracking*

The first experiment tests whether feedback alone can maintain tracking accuracy when the reference trajectory contains strong manoeuvring dynamics. The complete proposed controller is compared with the proposed controller without flatness feedforward, in which the differential-flatness feedforward channel is removed, to evaluate the role of flatness feedforward in highly dynamic reference tracking. The

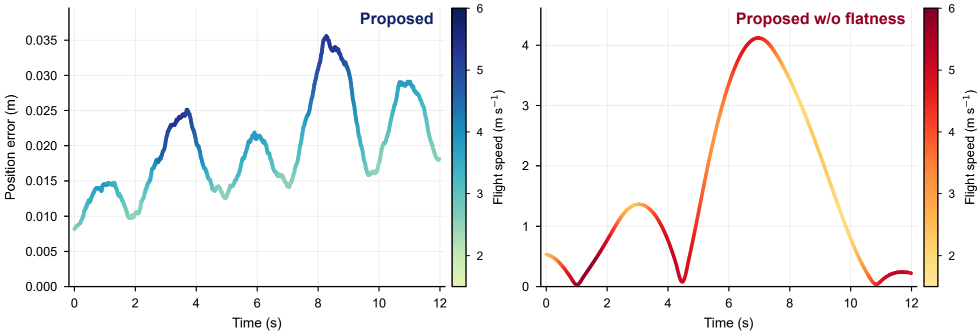

Figure 3. *Position error in highly dynamic figure-eight trajectory tracking.*

reference trajectory is a figure-eight curve:

$$
\begin{aligned}
x_r(t) &= A_x \sin\left(\frac{2\pi}{T}t\right), \\
y_r(t) &= A_y \sin\left(\frac{4\pi}{T}t\right), \\
z_r(t) &= z_0, \qquad \psi_r(t) = \psi_0,
\end{aligned}
\tag{65}
$$

where $A_x = 6\ \mathrm{m}$, $A_y = 3\ \mathrm{m}$, $T = 10\ \mathrm{s}$ and $z_0 = -1\ \mathrm{m}$, with $\psi_0 = 0$.

Figure 3 compares the error evolution of the complete controller and the controller without differential-flatness feedforward. Curve colour denotes the flight speed at the corresponding time, highlighting the relationship between speed variation and tracking error. The complete controller maintains centimetre-level tracking accuracy in the stable window, with RMSE and maximum error values of $0.021\ \mathrm{m}$ and $0.040\ \mathrm{m}$, respectively. After flatness feedforward is removed, the error is amplified in high-speed segments and segments with rapidly changing curvature, and the RMSE and maximum error increase to $2.045\ \mathrm{m}$ and $4.122\ \mathrm{m}$, respectively.

Relative to the version without flatness feedforward, the complete controller reduces RMSE by approximately 98.97% and maximum error by approximately 99.03%. This difference indicates that differential-flatness feedforward accounts for the predictable acceleration, thrust-direction and attitude-variation demands in the reference trajectory, allowing the feedback terms to mainly correct modelling errors and discrete-sampling residuals. Without this feedforward channel, the feedback loop must both follow the reference dynamics and correct tracking errors, which more readily produces lag in high-speed segments or where the trajectory bends rapidly. The decrease in the error curve without flatness feedforward does not indicate recovery of stable tracking. It is caused by the periodicity of the figure-eight reference trajectory, because the geometric distance can temporarily decrease when the reference point returns close to the lagging UAV.

This ablation experiment supports embedding flatness feedforward in the disturbance-rejection control framework. In continuous turning and rapid path-adjustment tasks for meteorological observation over hilly terrain, relying only on feedback correction converts predictable trajectory dynamics into an error-compensation problem. Feedforward provides the main control demand in advance and reduces the burden on the feedback channel. Because this experiment does not include external wind disturbances, its conclusion is mainly limited to highly dynamic reference-tracking capability. The next experiment further evaluates disturbance rejection under a synthetic spatiotemporal wind-force field.

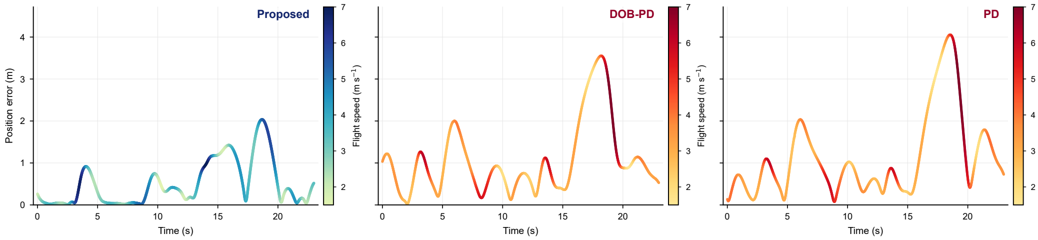

Figure 4. *Trajectory-tracking position error under the synthetic spatiotemporal wind-force field.*

6.3. Tracking under spatiotemporal wind

The second experiment considers manoeuvring tracking under spatiotemporally varying wind disturbances. It examines whether disturbance-observer compensation alone is sufficient, or whether the feedforward, feedback and compensation structures must be coordinated. PD, DOB-PD and the proposed controller are compared under a synthetic spatiotemporal wind-force field. PD contains no explicit disturbance compensation, whereas DOB-PD adds disturbance-observer compensation to the same linear feedback structure. For a fair comparison, all three controllers use differential-flatness feedforward.

All three controllers track the figure-eight trajectory in (65) and use the same initial position, sensor noise, actuator constraints and synthetic spatiotemporal wind-disturbance input. The PD controller uses linear position and attitude feedback. DOB-PD adds the same type of smooth bounded disturbance-observer compensation as the proposed controller to the linear feedback structure. The proposed controller combines differential-flatness feedforward, $\tanh(\cdot)$-based smooth bounded feedback and smooth bounded disturbance-observer compensation.

The synthetic spatiotemporal wind-force field is injected in MuJoCo as an external force acting on the vehicle centre of mass. Let the UAV position in the NED coordinate system be $\boldsymbol{\xi} = [x, y, z]^T$. The wind-disturbance force is

$$\boldsymbol{f}_w(\boldsymbol{\xi}, t) = \boldsymbol{f}_0 + \boldsymbol{a}_g \sin\left(2\pi f_g t + \boldsymbol{k}_s^T \boldsymbol{\xi}\right) + \boldsymbol{f}_{\mathrm{vor}}(\boldsymbol{\xi}), \tag{66}$$

where $\boldsymbol{f}_0 = [0, 0, 0]^T$ N is the mean wind disturbance, $\boldsymbol{a}_g = [2.0, 1.0, 0.4]^T$ N is the gust amplitude, $f_g = 0.08$ Hz is the temporal frequency and $\boldsymbol{k}_s = [0.7, 0.5, 0]^T \ \mathrm{rad\,m^{-1}}$ is the spatial frequency. The local vortex term is

$$\boldsymbol{f}_{\mathrm{vor}}(\boldsymbol{\xi}) = F_{\mathrm{vor}} \exp\left[-\left(\frac{\rho}{R_{\mathrm{vor}}}\right)^2\right] \begin{bmatrix} -\rho_y/\rho \\ \rho_x/\rho \\ 0 \end{bmatrix}, \tag{67}$$

where $[\rho_x, \rho_y]^T = [x - c_x, y - c_y]^T$, $\rho = \sqrt{\rho_x^2 + \rho_y^2}$, the vortex centre is $\boldsymbol{c} = [0, 0, -1]^T$ m, the vortex radius is $R_{\mathrm{vor}} = 4.0$ m and the vortex-force magnitude is $F_{\mathrm{vor}} = 1.5$ N. When ρ approaches zero, the vortex term is set to zero to avoid a directional singularity.

Figure 4 shows the position errors of the three controllers within the stable window under the synthetic spatiotemporal wind-force field. PD and DOB-PD both show metre-level peaks when trajectory manoeuvring and local vortex disturbance are superimposed, with maximum error values of 4.058 m and 3.588 m, respectively. The proposed controller achieves RMSE and maximum error values of 0.735 m and 2.039 m, respectively, compared with 1.366 m and 3.588 m for DOB-PD and 1.503 m and 4.058 m for PD. Relative to DOB-PD, the proposed controller reduces RMSE and maximum error by approximately 46.19% and 43.17%, respectively. Relative to PD, the reductions are approximately 51.10% and 49.75%, respectively.

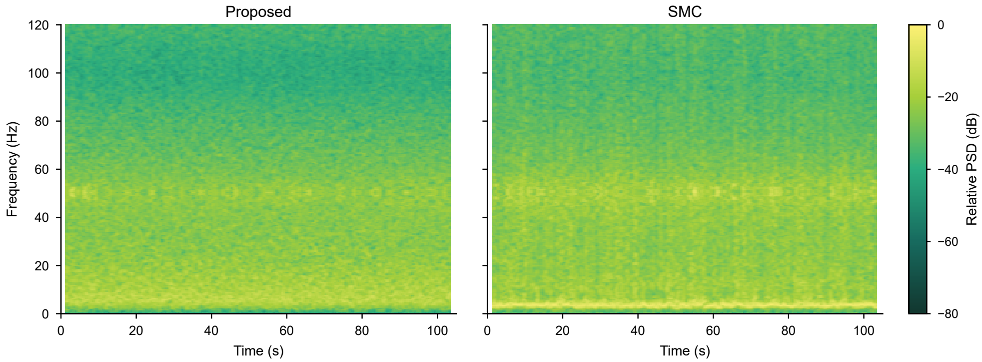

Figure 5. *Time-frequency PSD of equivalent motor-thrust commands during hovering.*

Adding disturbance-observer compensation to the PD structure improves part of the disturbance response, but it does not eliminate peak errors caused by the combined action of the synthetic spatiotemporal wind-force field and the manoeuvring trajectory. The error peaks of PD and DOB-PD have similar temporal structures, indicating that the error is not caused only by the external wind-disturbance input, but is also related to reference dynamics, underactuated attitude-thrust coupling and input constraints. The proposed controller uses flatness feedforward, smooth bounded feedback and disturbance-observer compensation simultaneously, assigning reference dynamics, unknown disturbances and residual error correction to different components. This coordinated structure gives lower RMSE and maximum error than the two comparison controllers.

In low-altitude mountain observation tasks, reducing the maximum trajectory error under spatiotemporally varying disturbances helps maintain observation routes and safety boundaries. The disturbance in this experiment is the superposition of a periodic-gust equivalent force and a local vortex force. It is a controlled synthetic external-force field rather than a complete model that maps wind speed to aerodynamic force. Therefore, this result mainly demonstrates the disturbance-rejection capability of the control structure under synthetic spatiotemporal wind-disturbance input. The observer gains in Table 2 were selected as implementation parameters rather than as a strict channel-by-channel verification of the sufficient inequalities in Proposition 1 and Corollary 1. Those inequalities provide conservative design guidance under bounded equivalent disturbances, while the simulations evaluate the resulting closed-loop behaviour under the specified force field, sensor noise and actuator limits. Measured wind fields or terrain-related flow fields still require further validation.

6.4. Input smoothness

The third experiment evaluates whether continuous bounded feedback can reduce broadband input oscillations caused by robust correction in a hovering task without external wind disturbance. SMC and the proposed controller are compared in terms of motor-input smoothness during wind-free hovering. SMC retains the robust switching term, whereas the proposed controller uses continuous $\mathtt{tanh}(\cdot)$ feedback and smooth bounded disturbance-observer compensation.

The simulation scenario is fixed-point hovering without an external wind field, and the reference position and reference yaw remain constant. This experiment uses the frequency-domain energy of the equivalent motor-thrust command as the evaluation metric. After subtracting the mean from each equivalent motor thrust, the short-time PSD is computed and averaged over the four motors, giving the time-frequency distribution shown in Fig. 5. Frequency-band energy values are obtained by integrating the linear PSD described in Section 6.1.

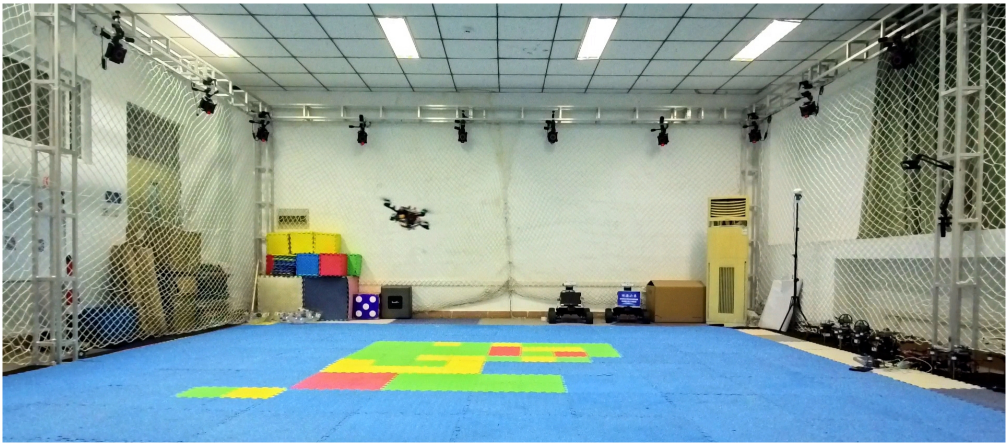

Figure 6. *Constructed quadrotor UAV flying in the motion-capture experimental site.*

The SMC motor inputs show stronger spectral energy from low to medium-high frequencies, especially with a more continuous high-PSD region over 0–60 Hz. By contrast, the proposed controller has lower overall spectral energy and weaker bright regions at high frequencies. Integrating the average PSD of the four motors over 0–120 Hz gives a band energy of 7.245×10^{-3} N^2 for the proposed controller and 1.423×10^{-2} N^2 for SMC, a reduction of approximately 49.09%. In the 20–120 Hz, 40–120 Hz and 80–120 Hz medium-high frequency bands, the integrated PSD values of the proposed controller are 2.927×10^{-3}, 1.680×10^{-3} and 1.229×10^{-4} N^2, respectively. Relative to the SMC values of 4.959×10^{-3}, 3.452×10^{-3} and 3.174×10^{-4} N^2, these correspond to reductions of approximately 40.98%, 51.33% and 61.27%.

This result is consistent with the structural difference between the two control laws. The switching term in sliding-mode control can improve robustness to uncertainty, but under the joint effects of discrete sampling, actuator allocation and numerical differentiation, it may introduce broadband oscillatory components into motor commands. The proposed controller designs the error correction as continuous $\tanh(\cdot)$-based smooth bounded feedback and supplements it with smooth bounded disturbance-observer compensation, thereby limiting the magnitudes of both feedback and compensation terms by gain upper bounds. This structure reduces the broadband energy of equivalent motor thrust in the hovering task.

On a real quadrotor platform, high-frequency motor-input energy may affect the thermal load of motors and electronic speed controllers, intensify propeller and frame vibration, and couple with inertial-sensor noise. This experiment indicates that, under wind-free hovering, the proposed smooth bounded control structure can weaken broadband motor-input oscillations. The present data do not show a single narrow-band structural resonance peak, so the conclusion should be interpreted as a reduction in broadband input energy rather than suppression of a specific mechanical resonance frequency. Hovering recovery under sudden gust disturbances and real-flight validation under different wind-field conditions remain topics for future work.

7. Real-flight validation

The real-flight experiment assesses whether the control framework shown in Fig. 1 can track a repeated-turn trajectory on the physical platform while keeping the position-tracking error within a stable range. A figure-eight flight experiment was conducted at the indoor site shown in Fig. 6. The parameters of the real quadrotor are consistent with the simulation settings in Table 1. The reference trajectory uses the horizontal path defined in (65), with $A_x = 1.5$ m, $A_y = 0.75$ m and $T = 15$ s, and with constant yaw.

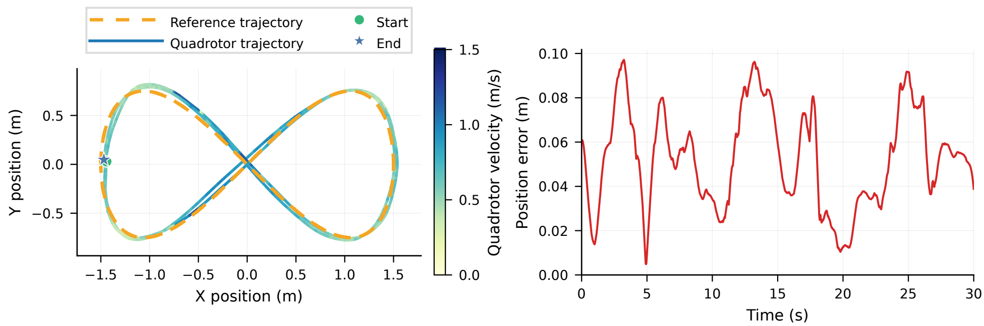

Figure 7. *Real-flight figure-eight trajectory validation: the left panel shows the reference trajectory and the measured quadrotor trajectory coloured by horizontal speed, and the right panel shows the position error relative to the reference trajectory.*

Figure 7 shows the reference trajectory, the measured quadrotor trajectory coloured by horizontal speed, and the position error. The measured trajectory has a horizontal span of $3.007\ \mathrm{m} \times 1.594\ \mathrm{m}$, an average flight height of $0.795\ \mathrm{m}$ and a height standard deviation of $0.016\ \mathrm{m}$. During this flight segment, the RMSE of the measured position relative to the reference trajectory is $0.057\ \mathrm{m}$, the maximum error is $0.097\ \mathrm{m}$ and the average horizontal speed is $0.616\ \mathrm{m/s}$. These results indicate that the proposed control framework can maintain centimetre-level root-mean-square error during repeated-turn trajectory tracking on the real platform, with maximum error below $0.10\ \mathrm{m}$.

Because of site and condition limitations, further real-flight experiments under complex terrain-related wind fields, different flight speeds and larger trajectories were not conducted. Nevertheless, validation on this representative trajectory supports the practical deployability of the proposed method.

8. Conclusion

This paper addressed the trade-off among manoeuvring trajectory tracking, bounded external disturbances and control-input smoothness in low-altitude observation over hilly terrain. It proposed a quadrotor control framework that combines differential-flatness feedforward, smooth bounded feedback and disturbance-observer compensation. The results indicate that assigning predictable reference dynamics to flatness feedforward, grouping wind-motivated force disturbances, model perturbations, unmodelled dynamics, virtual-control derivative terms and feedforward residuals into an equivalent total disturbance, and limiting error-correction and compensation magnitudes with $\tanh(\cdot)$-type continuous bounded feedback can reduce tracking error and broadband motor-input energy in the tested settings.

Three simulation experiments and one real-flight trajectory validation support this conclusion from different perspectives. The complete controller maintains centimetre-level error in highly dynamic figure-eight tracking and reduces RMSE and maximum error relative to the version without flatness feedforward, indicating that the feedforward channel accounts for the main reference dynamics. Under the synthetic spatiotemporal wind-force field, the RMSE and maximum error of the proposed controller are lower than those of PD and DOB-PD, indicating that smooth bounded feedback and disturbance-observer compensation are more effective when combined than when observer compensation is added only to linear feedback. In wind-free hovering, the proposed controller reduces the integrated PSD of equivalent motor thrust over 0–120 Hz relative to SMC, suggesting that the structure helps weaken broadband motor-input oscillations. In real-flight figure-eight validation, the measured trajectory has an RMSE of $0.057\ \mathrm{m}$ relative to the reference trajectory, indicating that the framework can execute repeated-turn trajectories on a real platform.

These results should be interpreted within the current experimental settings. The second experiment uses a synthetic external-force field acting on the vehicle centre of mass rather than a complete model that maps wind speed to aerodynamic force. The third input-smoothness experiment covers only wind-free hovering. The real-flight validation includes only one stable figure-eight trajectory. The results therefore support the effectiveness of the proposed structure in controlled MuJoCo scenarios and in representative real-flight trajectory validation, but they do not constitute a performance guarantee in real complex-terrain wind fields or across multiple real-platform operating conditions. Future work will combine measured wind fields, terrain-related flow fields and real-platform tests to examine the applicability range of the framework.

Author Contributions. First Author conceived and designed the study, developed the controller, conducted the simulations and real-flight experiments, analysed the data and wrote the manuscript. All authors reviewed and approved the final manuscript.

Financial Support. This research received no specific grant from any funding agency, commercial or not-for-profit sectors.

Conflicts of Interest. The authors declare no conflicts of interest exist.

Ethical Approval. Not applicable.

Data Availability Statement. The code and simulation scripts supporting this study are available at https://github.com/SWUST-ICAA/tanh-ctrl.

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